PalmSource™ 99

Strategies for Debugging Palm OS® Applications

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Session Agenda

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- Resetting your Palm™ device
- Handy shortcuts
- Using POSE, the Palm OS Emulator
- Code-level debugging
- Codewarrior Debugging Console
- Walkthrough
- Q&A

Resetting Your Palm™ Device

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Soft-reset

- Press reset button on rear of device
- Resets dynamic heaps, but leaves storage heaps untouched
  - OS restarts with new stack, new global variables, restarted drivers and a reset serial port
- Each installed application receives a `sysAppLaunchCmdSystemReset` launch code

Resetting Your Palm™ Device

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No-notify reset

- Press reset button on rear of device while holding down the up-arrow
- Similar to soft reset except the `sysAppLaunchCmdSystemReset` launch code is not sent to each application
  - No system patches are loaded

Resetting Your Palm™ Device

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Debug reset

- Press reset button on rear of device while holding down the down-arrow
- Flashing box in upper-left corner indicates device is in debugger mode
  - The serial port is open, waiting for a debugger to connect

Resetting Your Palm™ Device

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Hard reset

- Press reset button on rear of device while holding down the power button
- Confirm hard reset by pressing the up-arrow when asked
- Same effect as soft reset, however all storage heaps are reset also
- If all hardware application buttons are also held down the device will not load default databases into RAM

Handy Shortcuts

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- Use Graffiti® function to enter shortcuts
  - Shortcut character (similar to lowercase l)
  - Followed by two taps (produces a period)
  - Then a specific numeric or alpha character

Handy Shortcuts

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- Shortcut…1
  - Places device in low-level debugger mode
  - Perform soft reset to exit
- Shortcut…2
  - Places device in high-level debugger mode
  - Device is waiting for debugger such as Codewarrior to connect
  - Perform soft reset to exit

Handy Shortcuts

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- Shortcut…3
  - Disables automatic power down
  - Perform soft reset to exit
- Shortcut…4
  - Displays the user name and HotSync® identifier used by HotSync® to identify unit

Handy Shortcuts

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- Shortcut…5
  - Erases user name and HotSync® id
    - Device will be treated as if it has never been synced before
- Shortcut…6
  - Displays the ROM built date for the device
- Shortcut…7
  - Switches battery profile between alkaline and NiCad
    - Not useful as profile apparently not completed for NiCads

Handy Shortcuts

- Shortcut…8
  - Toggles backlight on Palm V device
- Shortcut…t
  - Toggles loopback test mode for Exchange Manager
    - Allows you to test app that uses Exchange Manager on a single device
- Shortcut…s
  - Toggles IR redirection between serial port and IR
- Shortcut…i
  - Initiates a beam receive

Palm OS® Emulator

- Emulates Palm™ devices at a hardware level
- Runs a standard (or debug) Palm OS ROM image
- Derived from Copilot by Greg Hewgill
- Source code is available
- Mac OS, Windows and Unix versions available

Palm OS® Emulator

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ROM images

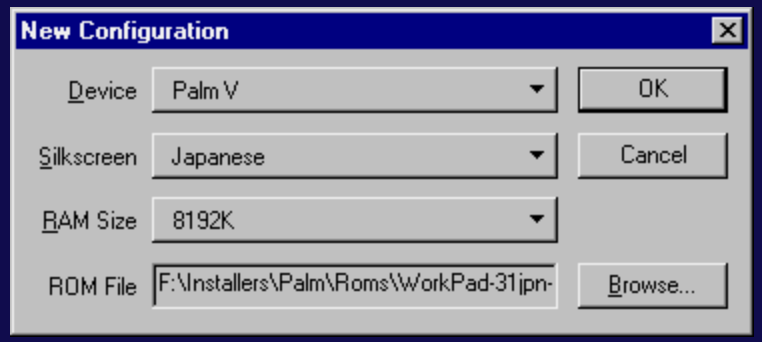
- ROM images transferred from real device
  - Use *Transfer ROM* option in POSE
- ROM images can be downloaded from www.palm.com
  - Both debug and non-debug ROMS are available
  - All versions of Palm devices including non-English such as German and Japanese

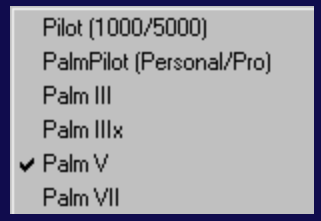
Palm OS® Emulator

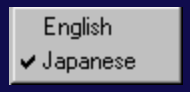
PalmSource 99

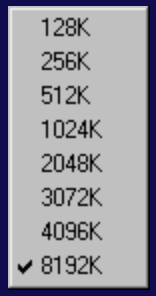
Multiple configurations

- You can create multiple configurations to test against
- Each configuration has a separate .psf file
- Caps-lock or New menu option to create new configuration





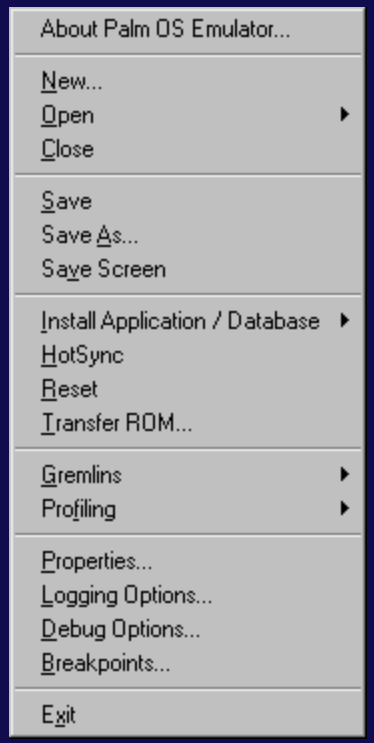




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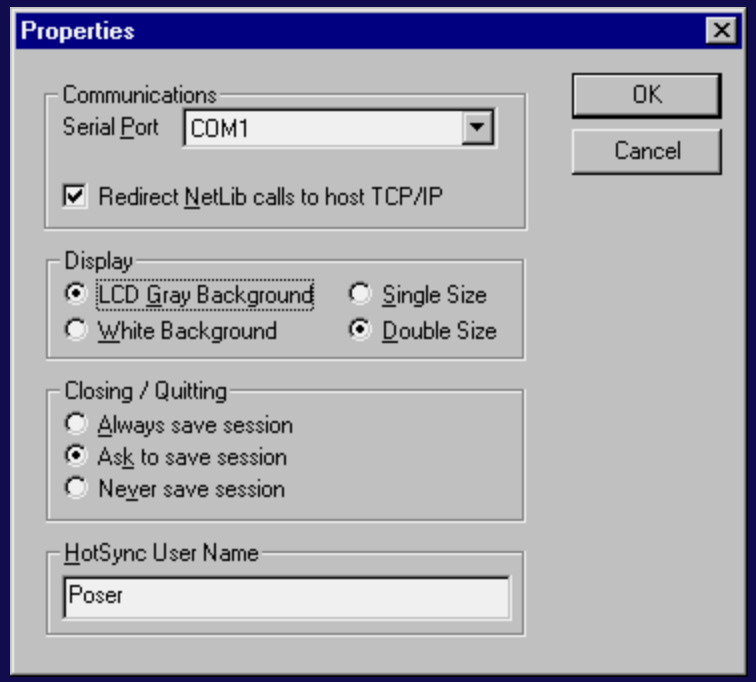
- A wealth of options under the menus



Palm OS® Emulator

Properties

- Redirect serial and NetLib activity
- Display options
- HotSync® user name

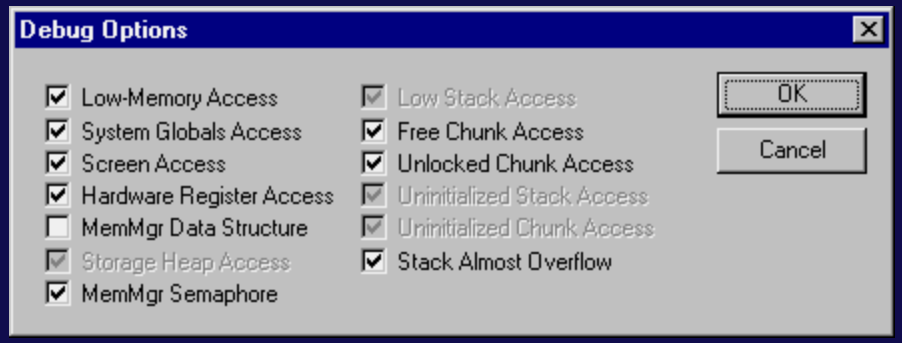


Palm OS® Emulator

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Debugging options

- POSE knows what your application is doing!
  - It emulates each memory access, so can trap any errant behavior

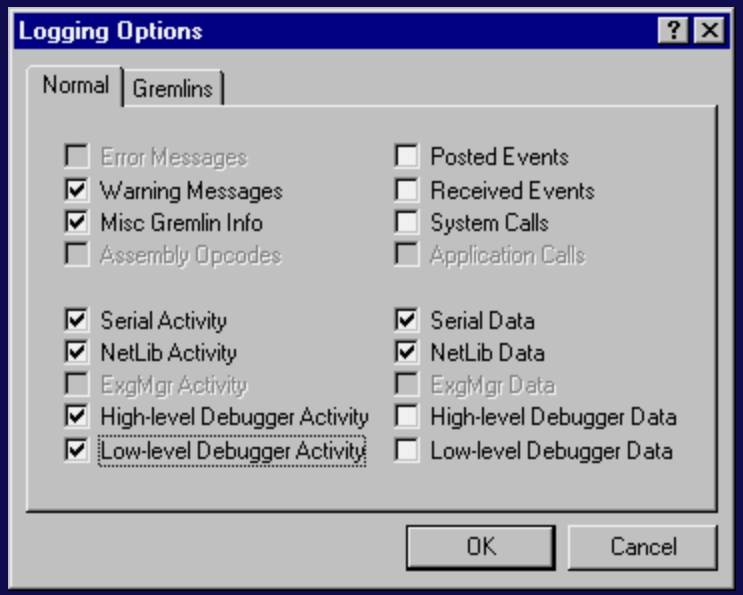


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Logging options

- Creates comprehensive log files

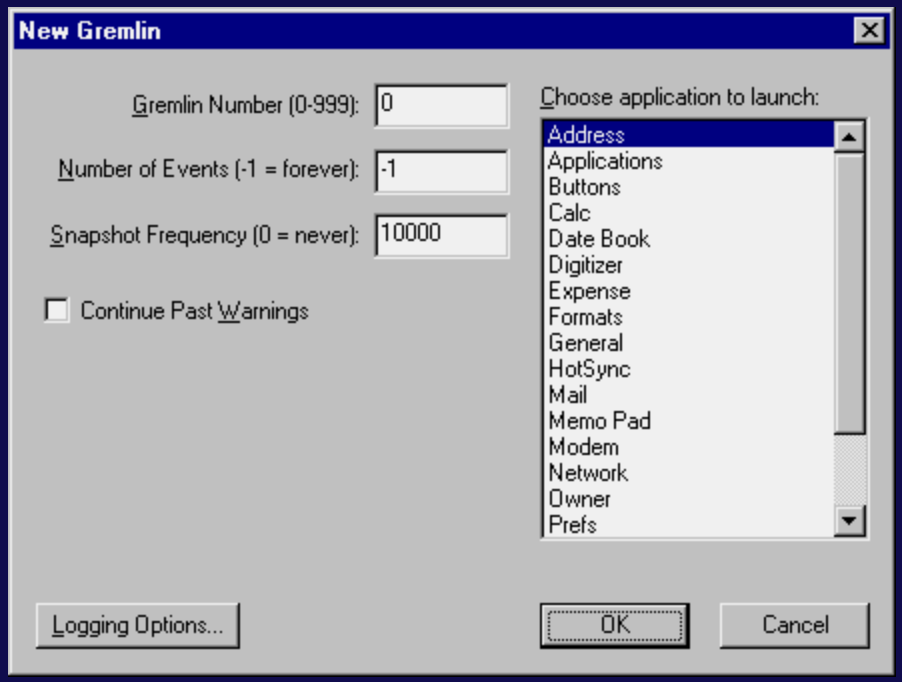


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Gremlins

- Mindless, repetitive testing

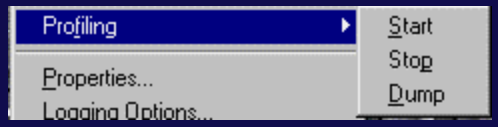


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Profiling

- Helps you find where your application spends most of its time



- Each function entry and exit is tracked
- Outputs for each function
  - Number of calls to it
  - Total, average and maximum time spent
  - Time spent by children

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Profiling

- Uses special version of POSE
- Applications must be built with MacsBug symbols (lets profiler see function names)
- Outputs a text file and a .MWP file that, currently, only MW Profiler under Mac OS can read

index	parent	depth	function name	count	only cycles	only msec	only %
43	-1	0	functions	0	0	0.000	0.0
51	43	1	FrmDispatchEvent	22	23976	1.000	0.0
52	51	2	WeekViewHandleEvent	22	37836	2.000	0.0
339	52	3	WeekViewGoToWeek	6	92472	5.000	0.0
424	339	4	WeekViewInit	6	154618	9.000	0.0
519	424	5	WeekViewSetTitle	6	21962	1.000	0.0
521	519	6	DateToAscii	7	29344	1.000	0.0
525	521	7	DmGetResource	7	18788	1.000	0.0
529	525	8	DmSearchResource	7	19824	1.000	0.0
538	529	9	DmFindResource	28	265090	15.000	0.0
539	538	10	PrvRsrcEntryPtr	1365	830564	50.000	0.0
540	539	11	unknown $10C14B62	1365	562380	33.000	0.0
544	529	9	MemSemaphoreRelease	7	2870	0.000	0.0

Code-Level Debugging

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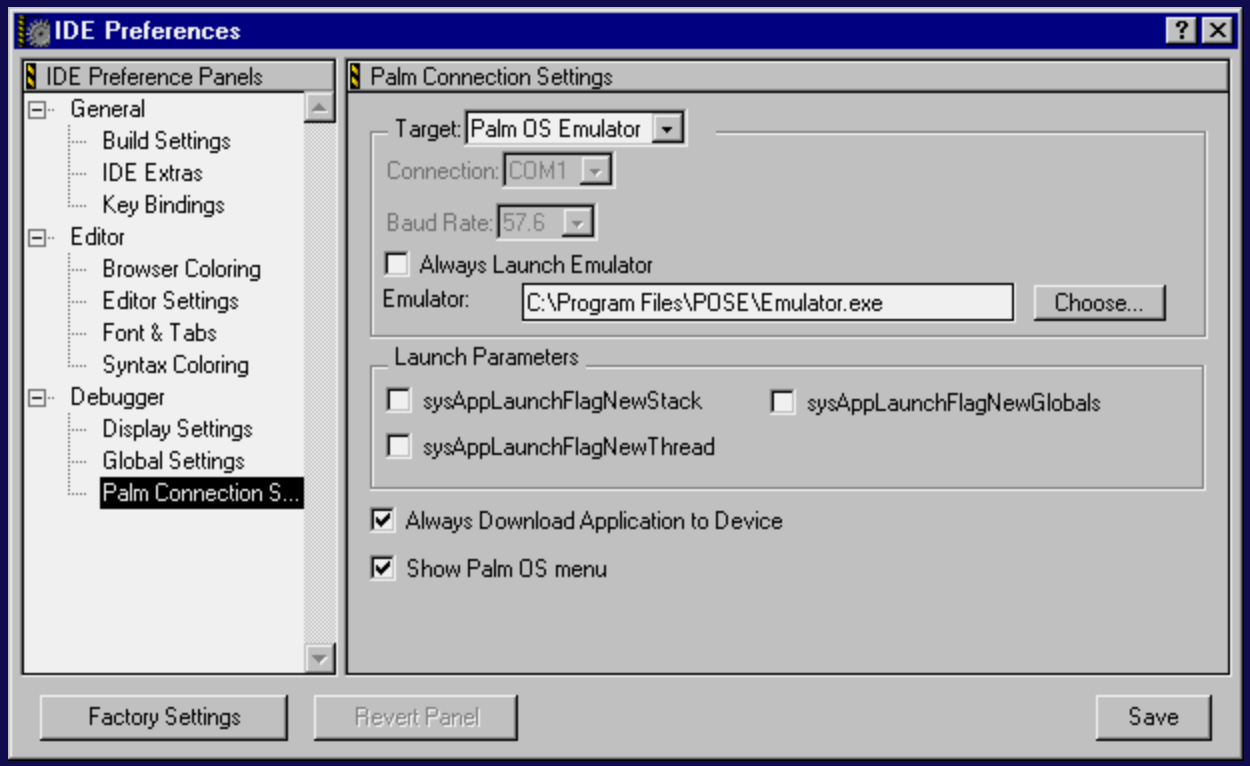
- Use either POSE or a real device
- Different IDE setups for each

Code-Level Debugging

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Debugging with POSE

- IDE preferences

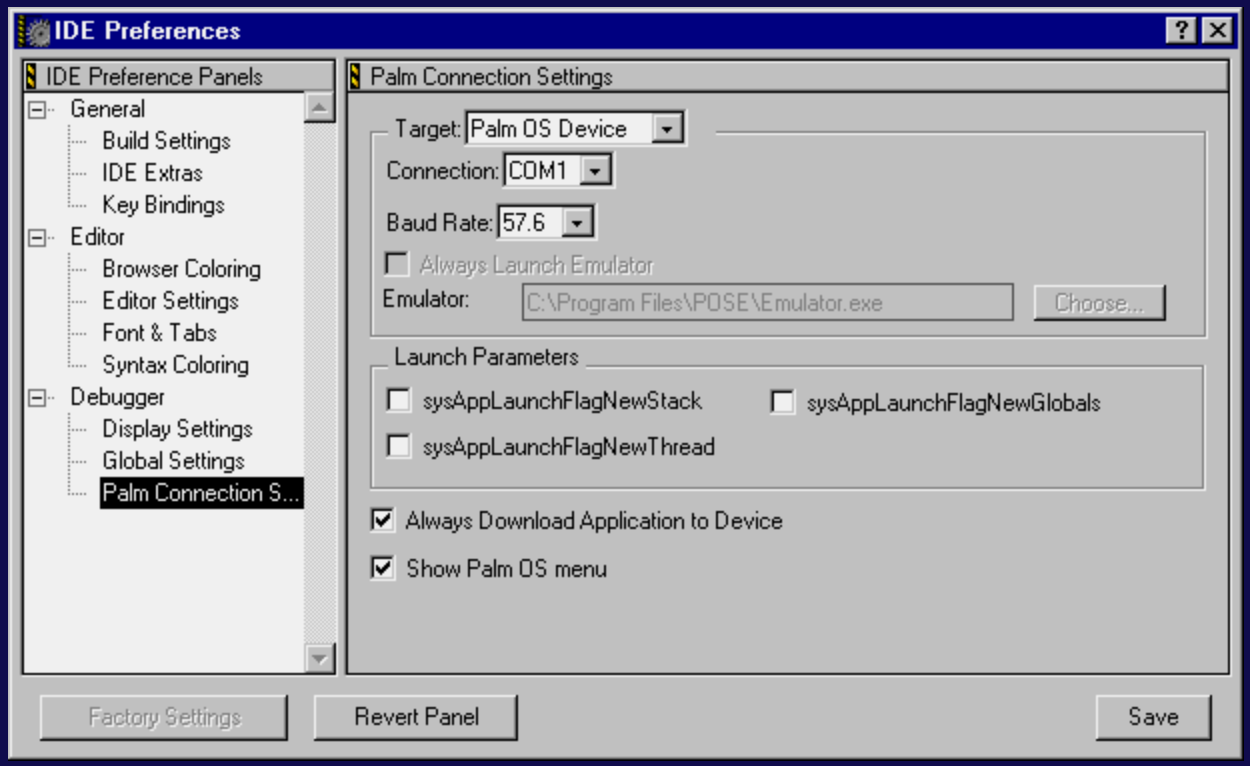


Code-Level Debugging

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Debugging with a real device

- IDE preferences

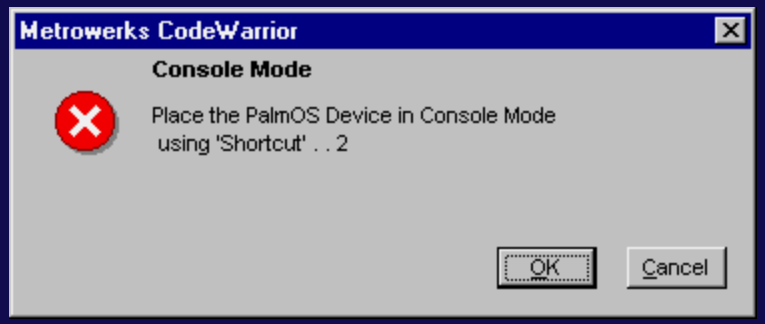


Code-Level Debugging

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Debugging with a real device

- After selecting Debug Codewarrior will prompt:

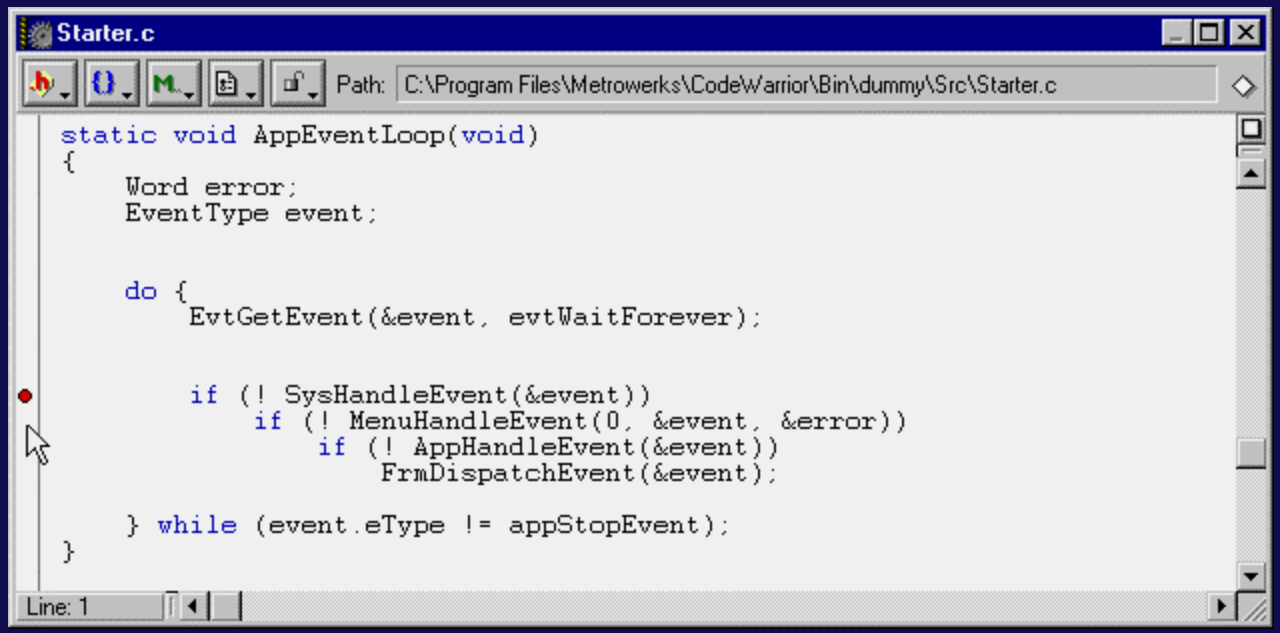


Code-Level Debugging

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Setting breakpoints

- Enable debugger
- Click in left-most column beside source line

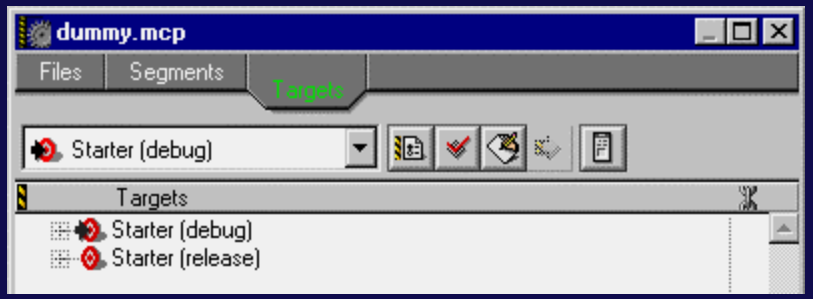


Debug vs Release Builds

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- Codewarrior allows you to create multiple targets
  - Typically used to create separate debug and release builds
  - Each target can have separate preference settings

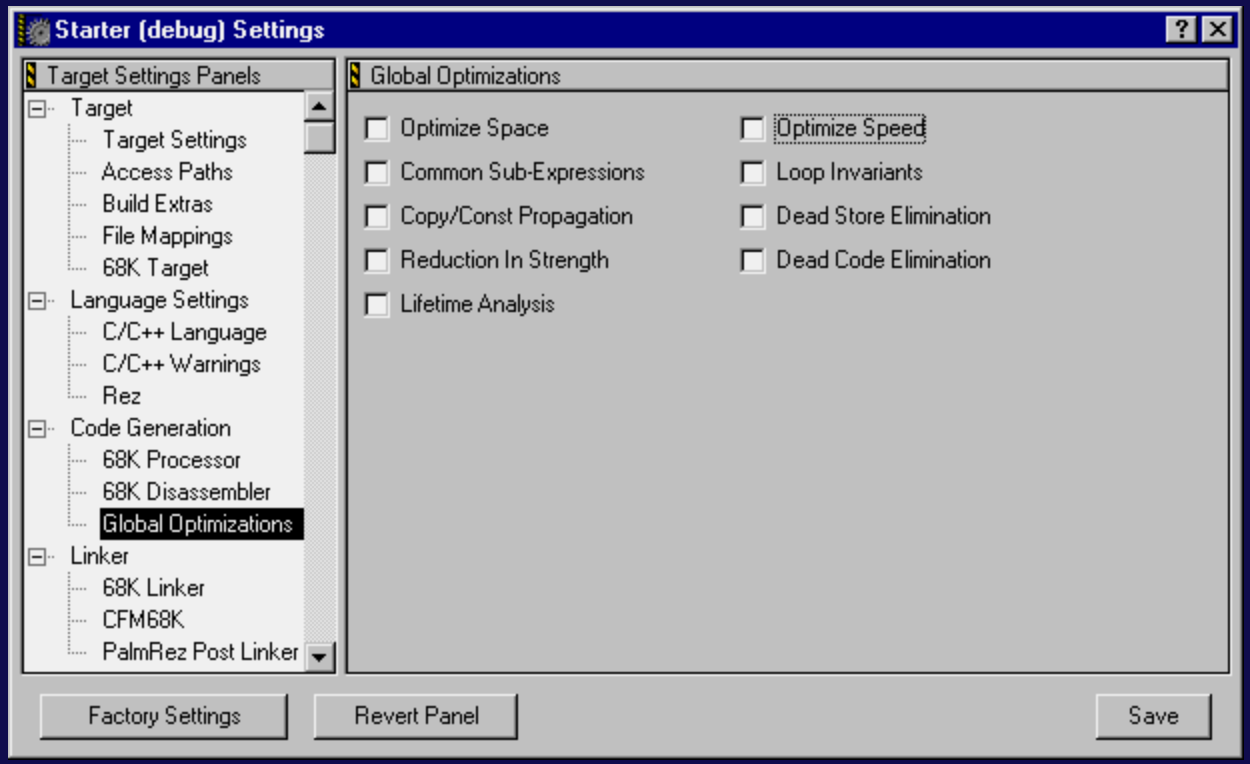




Debug vs Release Builds

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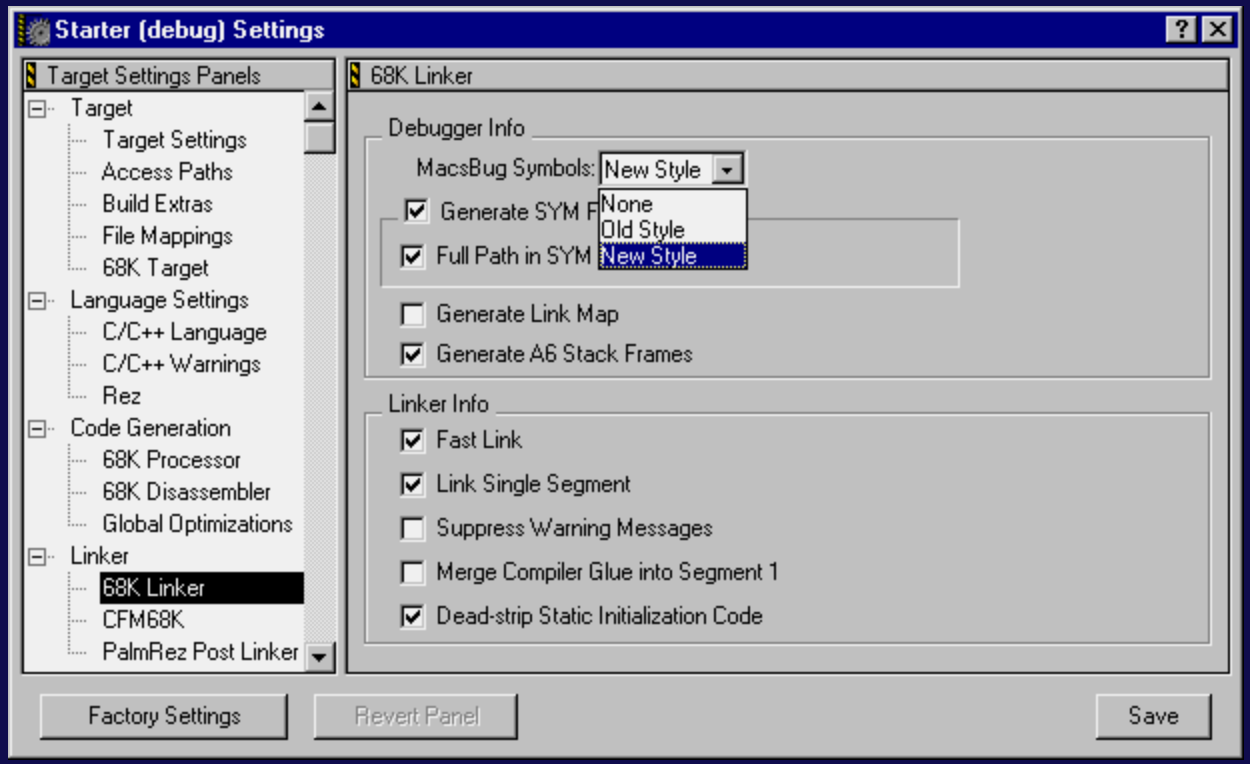
- Optimizations:



Debug vs Release Builds

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- Embedded function names:



Debugging Console

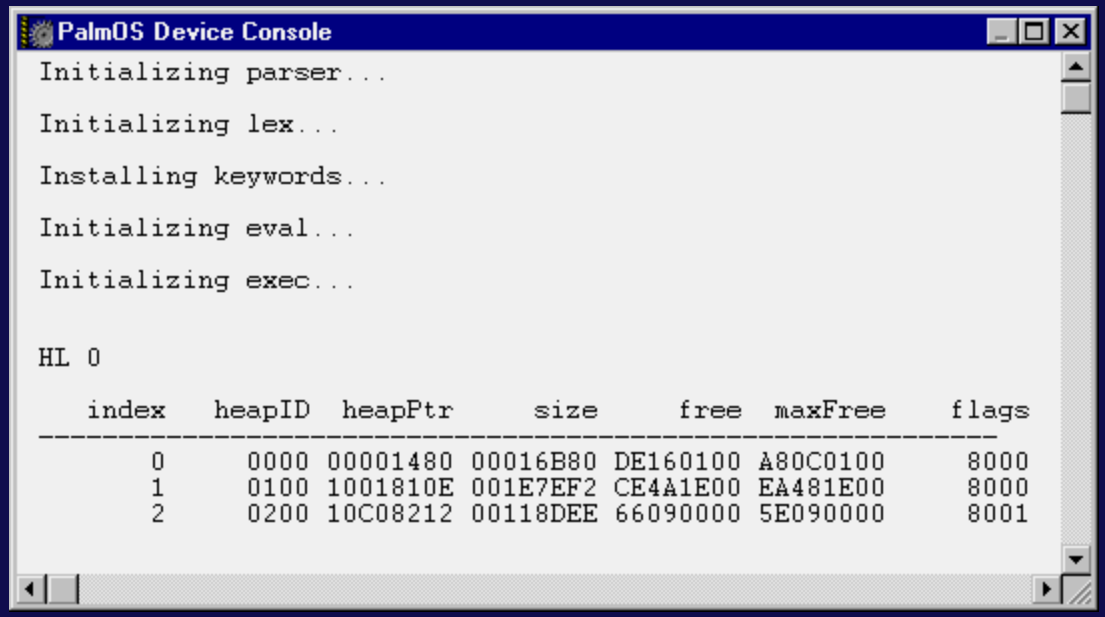
- Command line interface to real device or POSE
- Inspect and modify memory, databases, features etc
- Dangerous in the wrong hands!
  - Backup your real device first!

Debugging Console

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Starting the console

- Select Open Debug Console from Palm OS menu in Codewarrior



Debugging Console

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Heap Utilities

- HL—list all heaps on a memory card
  - **`hl <card number>`**
- HD—dump heap contents
  - **`hd <heap id>`**
- HT—display heap summary information
  - **`ht <heap id>`**
- HC—compacts a heap
  - **`hc <heap id>`**

Debugging Console

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Heap Utilities

- HChk—checks heap integrity
  - **`hchk <heap id>`**
- HS—scrambles a heap
  - **`hs <heap id>`**
- HF—fills a heap with memory chunks
  - **`hf <heap id> [freeBytes]`**

Debugging Console

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Database Utilities

- Import—imports a database
  - **`import <card no> <filename>`**
- Export—exports a database
  - **`export <card no> <filename>`**
- Create—creates a new database
  - **`create <card no> <name> [options…]`**
    - **`t <type>`**
    - **`c <creator>`**
- Del—deletes a database
  - **`del <card no> <databaseName>`**

Debugging Console

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Database Utilities

- Dir—lists database directory
  - **`dir <card no>|<searchOptions>`**
    - **`t <type>`**
    - **`c <creator>`**
- Open—opens a database
  - **`open <card no> <name>`**
- Close—closes a database
  - **`close <access ptr>`**
- Opened—list all opened databases
  - **`open`**

Debugging Console

Record Utilities

- AddRecord—adds a record to a database
  - **`addrecord <access ptr> <index> <record text>`**
- DelRecord—deletes a record
  - **`delrecord <access ptr> <index>`**
- ChangeRecord—replaces record in a database
  - **`changerecord <access ptr> <index> <record text>`**
- ListRecords—displays records in a database
  - **`listrecords <access ptr>`**

Walkthrough


Q&A


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